

$$5.6 \quad df = 63 - 3 = 60$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

$$\widehat{\text{cov}}(b_1, b_2, b_3) = \begin{matrix} & \begin{matrix} b_1 & b_2 & b_3 \end{matrix} \\ \begin{matrix} b_1 \\ b_2 \\ b_3 \end{matrix} & \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \end{matrix}$$

5% significance level

a) 1) $H_0 : B_2 = 0$
 $H_1 : B_2 \neq 0$

2) Set $\alpha = 0.05$

3) Test statistic its sampling distribution:

$$t_{\text{stat}} = \frac{b_2 - 0}{\text{se}(b_2)} = \frac{3 - 0}{\sqrt{4}} = 1.5$$

$t_{0.975, 60} = 2$. The rejection region is $\{t: |t| > 2\}$.

$\Rightarrow$ We do not reject $H_0 : B_2 = 0$

b) 1) $H_0 : B_1 + 2B_2 = 5$
 $H_1 : B_1 + 2B_2 \neq 5$

2) $\alpha = 0.05$

3)
$$t_{\text{stat}} = \frac{(b_1 + 2b_2) - 5}{\text{se}(b_1 + 2b_2)} = \frac{(2 + 2 \cdot 3) - 5}{\sqrt{3 + 4 \cdot 4 + 2 \cdot 2 \cdot (-2)}} \approx 0.5045$$

$t_{0.975, 60} = 2$. The rejection region is $\{t: |t| > 2\}$.

$\Rightarrow$ We do not reject $H_0 : B_1 + 2B_2 = 5$

c) 1) $H_0 : B_1 - B_2 + B_3 = 4$
 $H_1 : B_1 - B_2 + B_3 \neq 4$

2) $\alpha = 0.05$

$$t_{\text{stat}} = \frac{b_1 - b_2 + b_3 - 4}{\text{se}(b_1 - b_2 + b_3)} = \frac{2 - 3 - 1 - 4}{\sqrt{3 + 4 + 3 - 2 \cdot (-2) + 2 \cdot 1 - 2 \cdot 0}} = -1.5$$

$t_{0.975, 60} = 2$. The rejection region is $\{t: |t| > 2\}$.

$\Rightarrow$ We do not reject $H_0 : B_1 - B_2 + B_3 = 4$

5.5
a)

$$x_t = t$$

From question, we have :

$$\begin{aligned}\sum (t - \bar{t})^2 &= \sum t^2 - \frac{(\sum t)^2}{T} \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4} \\ &= T(T+1) \left[\frac{2T+1}{6} - \frac{T+1}{4} \right] \\ &= T(T+1) \cdot \frac{4T+2-3T-3}{12} \\ &= T(T+1) \cdot \frac{T-1}{12} = \frac{T(T^2-1)}{12}\end{aligned}$$

$$\widehat{\text{Var}}(b_2) = \frac{\hat{\sigma}^2}{\sum (b_i - \bar{x})^2} = \frac{12 \cdot \hat{\sigma}^2}{T(T^2-1)}$$

$$\lim_{T \rightarrow \infty} \text{var}(b_2|x) = \lim_{T \rightarrow \infty} \frac{12 \hat{\sigma}^2}{T(T^2-1)} = 0$$

$\Rightarrow \text{var}(b_2|x) \rightarrow 0 \Rightarrow b_2$ is consistent.

b) $\sum_{t=1}^T (x_t - \bar{x})^2 = \sum_{t=1}^T x_t^2 - \frac{(\sum_{t=1}^T x_t)^2}{T}$; $x_t = (0,5)^t \Rightarrow \sum_{t=1}^T x_t = \sum_{t=1}^T (0,5)^t$

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1-r^n)}{1-r} \quad \text{with } r = 0,5$$

$$\Rightarrow \sum_{t=1}^T (0,5)^t = \frac{0,5(1-0,5^T)}{1-0,5} = 1 - 0,5^T$$

We have : $x_t^2 = (0,5^t)^2 = 0,25^t$

$$\Rightarrow \sum x_t^2 = \sum 0,25^t = \frac{0,25(1-0,25^T)}{1-0,25} = \frac{1}{3} [1 - 0,25^T]$$

$$\begin{aligned}\sum (x_t - \bar{x})^2 &= \sum x_t^2 - \frac{(\sum x_t)^2}{T} \\ &= \frac{1-0,25^T}{3} - \frac{(1-0,5^T)^2}{T}\end{aligned}$$

$$\widehat{\text{Var}}(b_2) = \frac{\hat{\sigma}^2}{\frac{1-0,25^T}{3} - \frac{(1-0,5^T)^2}{T}} \left\{ \begin{array}{l} 0,25^T \rightarrow 0 \Rightarrow \frac{1-0,25^T}{3} \rightarrow \frac{1}{3} \\ 0,5^T \rightarrow 0 \Rightarrow 1 - \frac{0,5^T}{T} \rightarrow 1 \end{array} \right\} \rightarrow \frac{(1-0,5^T)^2}{T} \rightarrow 0$$

$$\Rightarrow \lim_{T \rightarrow \infty} \text{var}(b_2) = \frac{\hat{\sigma}^2}{\frac{1}{3} - 0} = 3\hat{\sigma}^2 \neq 0 \Rightarrow b_2 \text{ is not consistent.}$$

c)

In (a), $x_t = t$ spread out as T increases $\rightarrow \sum (x_t - \bar{x})^2$ increase to infinity $\rightarrow \text{var} \rightarrow 0 \rightarrow b_2$ is consistent

$$\text{In (b), } \sum (x_t - \bar{x})^2 = \frac{1 - 0.25^t}{3} - \frac{(1 - 0.5^t)^2}{T};$$

when T increase, $\sum (x_t - \bar{x})^2 \rightarrow \frac{1}{3} = \text{constant}$;
so $\text{var}(b_2) \not\rightarrow 0 \rightarrow b_2$ is not consistent.

(c) In part (a), increasing T provides more and more information about β_2 and the relationship between y and x . However, in part (b), $x_t = 0.5^t$ becomes zero for moderately sized T , and increasing T further does not provide any more information about β_2 .

5.20

$$a) \bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1$$

$$\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$$

$$\begin{aligned} \hat{\beta}_{2, \text{mean}} &= \frac{(\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2) - (\beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} \\ &= \frac{\beta_2 \bar{x}_2 - \beta_2 \bar{x}_1 + \bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \\ &= \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \end{aligned}$$

$$E(\hat{\beta}_{2, \text{mean}} | x) = \beta_2 + \frac{E(e_2 | x) - E(e_1 | x)}{\bar{x}_2 - \bar{x}_1} = \beta_2 \quad (\text{Because } E(e_i | x) = 0, \text{ according to MLR})$$

$\Rightarrow \hat{\beta}_{2, \text{mean}}$ is unbiased of β_2

$$b) \hat{\beta}_2 = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \quad (\beta_2 = \text{const})$$

$$\begin{aligned} \Rightarrow \text{var}(\hat{\beta}_2 | x) &= \text{var}\left(\frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \mid x\right) = \frac{\text{var}(e_2 | x) + \text{var}(e_1 | x) - 2\text{cov}(e_1, e_2 | x)}{(\bar{x}_2 - \bar{x}_1)^2} \\ &= \frac{\frac{\sigma^2}{N} + \frac{\sigma^2}{N}}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \end{aligned}$$

$$(\text{Because : } \text{var}(e_i) = \sigma^2 \Rightarrow \text{var}(e_2 | x) = \text{var}(e_1 | x) = \frac{\sigma^2}{N} = \frac{2\sigma^2}{N}; \text{cov}(e_i, e_j) = 0)$$

$$\Rightarrow \text{var}(\hat{\beta}_2 | x) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

c)

$$\text{var}(\hat{\beta}_{2, \text{mean}}) = E[\text{var}(\hat{\beta}_{2, \text{mean}} | x)] + \text{var}[E(\hat{\beta}_{2, \text{mean}} | x)]$$

We have : $E(\hat{\beta}_{2, \text{mean}} | x) = \beta_2 = \text{const.}$ (from ques. a)

$$\Rightarrow \text{var}[E(\hat{\beta}_{2, \text{mean}} | x)] = 0$$

$$\text{var}(\hat{\beta}_{2, \text{mean}}) = E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] = \frac{4\sigma^2}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right]$$

$$d) \text{var}(\hat{\beta}_{2, \text{mean}} | x) = \frac{4b^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

$$\text{Condition: } N \rightarrow \infty \Rightarrow N(\bar{x}_2 - \bar{x}_1)^2 \rightarrow \infty$$

$$\Rightarrow (\bar{x}_2 - \bar{x}_1) = \text{const} (\neq 0)$$

$\Rightarrow \hat{\beta}_{2, \text{mean}}$ is a consistent estimator if $x_2 - x_1$ has value that's different from zero.

f) 1) $\text{var}(b_{2, \text{OLS}}) < \text{var}(\hat{\beta}_{2, \text{mean}})$ always, aligns with Gauss-Markov Theorem, which states that OLS has smallest variance in all of the Linear Unbiased Estimator.

2) As N increases, the variance of both gradually decreases and approaches toward zero, indicating that $\hat{\beta}_{2, \text{mean}}$ is consistent.

$$g) E[(s_x^2)^{-1}] \rightarrow 0,12 \text{ and } E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \rightarrow 0,16 \text{ with } x \sim U(0,10)$$

$$E[(s_x^2)^{-1}] \rightarrow 0,12 \text{ with } x \sim U(0,10)$$

$$\text{Var}(X) = \frac{(10-0)^2}{12} = \frac{100}{12} = 8,333$$

$$s_x^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 \xrightarrow{p} \text{Var}(X) = \frac{100}{12}$$

$$\frac{1}{s_x^2} = \frac{12}{100} = 0,12$$

h) If the obs are randomly assigned without ordering, both halves become random samples from the population. By LLN, both $\bar{x}_1$ and $\bar{x}_2$ converge to the population mean $E(x)$

$$\bar{x}_1 \xrightarrow{p} E(x) \quad \bar{x}_2 \xrightarrow{p} E(x)$$

$$\text{Then } (\bar{x}_2 - \bar{x}_1) \xrightarrow{p} 0, \frac{4b^2}{N(\bar{x}_2 - \bar{x}_1)^2} \rightarrow \infty$$

The estimator would be inconsistent, failing to capture the variation in x

$$E\left(\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right) \xrightarrow{p} 0,16$$

$$\bar{x}_1 \xrightarrow{p} E[U(0,5)] = 2,5$$

$$\bar{x}_2 \xrightarrow{p} E[U(5,10)] = 7,5$$

$$(\bar{x}_2 - \bar{x}_1) \xrightarrow{p} 7,5 - 2,5 = 5$$

$$\frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \xrightarrow{p} \frac{4}{5^2} = \frac{4}{25} = 0,16$$